Powers and roots are more important operations than multiplication and division. They must be done first.

Work out $\square^2$ $\sqrt{\ }$ before $\times$ $\div$ and then do $+$ $-$

(i) $24 - \sqrt{25} + 3^2$
$= 24 - 5 + 9$
$= 19 + 9$
$= 28$

(ii) $63 \div 9 + \sqrt{36}$
$= 63 \div 9 + 6$
$= 7 + 6$
$= 13$

If there are **brackets** in a calculation, do the calculation inside them first.

When a calculation involves brackets as well as other operations, this is the correct order to work out the answers.

First	Second	Third	Fourth
( )	$\square^2$ or $\sqrt{\ }$	$\times$ or $\div$	$+$ or $-$

B – brackets
I – indices (powers)
R – roots
D – division
M – multiplication
A – addition
S – subtraction

Example 1

Work out (i) $12 + 8 \div 2 + 4 \times 3$ (ii) $7 \times 8 - 3 - (72 \div 8)$

(i) $12 + 8 \div 2 + 4 \times 3$
$= 12 + 4 + 12$
$= 28$

(ii) $7 \times 8 - 3 - (72 \div 8)$
$= 7 \times 8 - 3 - (9)$
$= 56 - 3 - 9$
$= 53 - 9$
$= 44$

Note: A horizontal line is the same as the $\div$ symbol.
For example, (i) $\frac{15}{3} = 15 \div 3 = 5$ (ii) $\frac{20 \div 5}{2} = \frac{4}{2} = 2$

Exercise 1.5

Work out numbers (1–15).

1. $5 + 3 \times 4$
2. $6 \times 2 + 5$
3. $9 \times 4 - 8$
4. $22 - 5 \times 4$
5. $18 \div 3 + 7$
6. $16 - 4 + 3 \times 9$
7. $5 + 16 \times 2 - 8$
8. $25 - 16 \div 4$
9. $5 \times 7 - 27 \div 3$
10. $6 \times 3 + 36 \div 4$
11. $5 \times 8 - 6 + 36 \div 3$
12. $15 - 10 \div 2 + 8$
13. $9 - 6 \div 2 + 5 \times 4$
14. $9 \times 6 - 27 \div 3$
15. $34 - 8 + 54 \div 9$

16. Copy and complete these calculations.
Each $\square$ stands for an operation ($+$, $-$, $\times$, or $\div$).

(i) $5 \square 7 \times 2 = 19$ (ii) $21 \square 3 + 11 = 18$ (iii) $18 + 6 \square 2 = 21$
(iv) $12 \square 2 \times 5 = 22$ (v) $15 \square 2 - 11 = 19$ (vi) $24 \square 8 \square 5 = 15$

17. Work out each of these:

(i) $4 \times (5 + 6)$ (ii) $7 \times (12 - 8)$ (iii) $9 \times (7 - 1)$
(iv) $(14 - 3) \times 2$ (v) $(6 - 3) \times 4$ (vi) $48 - 3 \times (9 - 3)$
(vii) $(36 - 12) \div 3 + 9$ (viii) $25 \div (8 - 3) - 2$ (ix) $5 \times 4 + 3 \times (10 - 4)$

18. Find the value of each of these:

(i) $15 - 10 \div 2 + 6 \times 4$ (ii) $5 \times 8 - 3 \times (7 - 4)$
(iii) $15 \div (7 - 4) + 12$ (iv) $(16 + 4) \div 5 + 14 \times 2$
(v) $2 \times (7 - 4) + 28 \div 7$ (vi) $(10 \times 3 - 20) \times 5$

19. Work out each of these:

(i) $\dfrac{22 + 13}{7}$ (ii) $\dfrac{5 + 9 - 2}{6}$ (iii) $\dfrac{100}{2 \times 5}$
(iv) $\dfrac{7 + 9}{10 - 6}$ (v) $26 - \dfrac{15}{3}$ (vi) $\dfrac{6 + 8 \times 3}{5}$

20. Find the value of each of these:

(i) $4 + 5^2$ (ii) $128 - 8^2$ (iii) $4 + 2 \times 3^2$
(iv) $2 \times \sqrt{25} + 4^2$ (v) $6 \times 3^2 - \sqrt{16}$ (vi) $2 \times 5^2 \div 10$

21. Work out each of these:

(i) $(3 + 5) \times 2 - 3^2$ (ii) $\sqrt{16} + 5 \times 2^2$ (iii) $4 \times \sqrt{9} + 6^2$
(iv) $4 \times 3^2 + 8 \div 4$ (v) $5^2 \times 2^2 - 8$ (vi) $2^2 + \sqrt{16} + 5^2$

22. Find the value of each of these:

(i) $\dfrac{8^2}{\sqrt{64}}$ (ii) $\dfrac{\sqrt{81} + 5}{4 \times 4 - 2}$ (iii) $\dfrac{2^2 \times \sqrt{100}}{4 \times (6 - 1)}$

23. Work out each of these:

(i) $3 + 5 \times 2^2 - 6 \times 3$ (ii) $(3 + 4)^2 - 2 \times 3^2$

24. Copy these statements and write in brackets to make them true.

(i) $5 \times 4 + 3 = 35$ (ii) $20 - 3 + 8 = 9$ (iii) $8 + 10 \div 2 = 9$
(iv) $48 \div 16 - 4 = 4$ (v) $11 - 6 \times 2 = 10$ (vi) $5 + 3 \times 4 - 2 = 30$

25. Which of the following has the greatest value? Find this value.

A: $(1 \times 2) \times (3 \times 4)$ B: $(1 \times 2) + (3 \times 4)$ C: $(1 \times 2) \times (3 + 4)$
D: $(1 + 2) \times (3 \times 4)$ E: $(1 + 2) \times (3 + 4)$